

Homework 7

15.2290.)

Notice that by definition, $\sigma_{\max}(A) = \|A\|$ where $\|A\|$ is the operator norm of A . We know that the operator norm of A is bounded above by the Frobenius norm of A . So if $|A_{ij}| \leq \epsilon$ for $1 \leq i, j \leq n$ then

$$\sigma_{\max}(A) = \|A\| \leq \|A\|_F = \left(\sum_{i=1}^n \sum_{j=1}^n |A_{ij}|^2 \right)^{\frac{1}{2}} \leq \left(\sum_{i=1}^n \sum_{j=1}^n \epsilon^2 \right)^{\frac{1}{2}} = (n^2 \epsilon^2)^{\frac{1}{2}} = n\epsilon$$

This upper bound is also tight. For example, if you have the following matrix and unit direction vector

$$E = \begin{bmatrix} \epsilon & \epsilon & \dots & \epsilon \\ \epsilon & \epsilon & \dots & \epsilon \\ \vdots & \vdots & \ddots & \vdots \\ \epsilon & \epsilon & \dots & \epsilon \end{bmatrix} \in \mathbb{R}^{n \times n}, \quad x = \begin{bmatrix} \frac{1}{\sqrt{n}} \\ \frac{1}{\sqrt{n}} \\ \vdots \\ \frac{1}{\sqrt{n}} \end{bmatrix} \in \mathbb{R}^n$$

Then

$$Ex = \begin{bmatrix} \epsilon & \epsilon & \dots & \epsilon \\ \epsilon & \epsilon & \dots & \epsilon \\ \vdots & \vdots & \ddots & \vdots \\ \epsilon & \epsilon & \dots & \epsilon \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{n}} \\ \frac{1}{\sqrt{n}} \\ \vdots \\ \frac{1}{\sqrt{n}} \end{bmatrix} = \begin{bmatrix} \sqrt{n}\epsilon \\ \sqrt{n}\epsilon \\ \vdots \\ \sqrt{n}\epsilon \end{bmatrix} = n\epsilon x \Rightarrow \sigma_{\max}(E) \geq \left\| \begin{bmatrix} \sqrt{n}\epsilon \\ \sqrt{n}\epsilon \\ \vdots \\ \sqrt{n}\epsilon \end{bmatrix} \right\| = \sqrt{n^2 \epsilon^2} = n\epsilon$$

Therefore, if $|A_{ij}| \leq \epsilon$ for $1 \leq i, j \leq n$ then $\sigma_{\max}(A) \leq n\epsilon$. This means that if a matrix has all small entries then it can't have an arbitrarily large gain in some direction, as the gain is bounded by the largest entry size and dimension of the matrix. It's worth noting that if you have control over the matrix dimensions, then you can make the gain as large as you want for fixed largest entry size ϵ . For instance, with our earlier example of E , if n is unbounded, then $\sigma_{\max}(E) \rightarrow \infty$ as $n \rightarrow \infty$ while keeping max entry size ϵ the same size. However, this apparent growth is due to increasing the matrix dimension rather than amplifying the effect of individual entries. When normalized by dimension, the gain remains small.

15.2300.)

a.

We want to prove that for $A \in \mathbb{R}^{n \times n}$,

$$\sqrt{\text{trace} A^T A} = \sqrt{\sum_{i=1}^n \sum_{j=1}^n |A_{ij}|^2}$$

If we define A by its columns, i.e. $A = [a_1 \ a_2 \ \dots \ a_n]$, then we can rewrite $A^T A$ as

$$A^T A = \begin{bmatrix} a_1^T \\ a_2^T \\ \vdots \\ a_n^T \end{bmatrix} [a_1 \ a_2 \ \dots \ a_n] = \begin{bmatrix} a_1^T a_1 & a_1^T a_2 & \dots & a_1^T a_n \\ a_2^T a_1 & a_2^T a_2 & \dots & a_2^T a_n \\ \vdots & \vdots & \ddots & \vdots \\ a_n^T a_1 & a_n^T a_2 & \dots & a_n^T a_n \end{bmatrix}$$

Therefore, we have

$$\text{trace} A^T A = a_1^T a_1 + a_2^T a_2 + \dots + a_n^T a_n = \sum_{j=1}^n a_j^T a_j = \sum_{j=1}^n \|a_j\|^2$$

Notice that for each a_j , we have

$$\|a_j\|^2 = |A_{1j}|^2 + |A_{2j}|^2 + \dots + |A_{nj}|^2 = \sum_{i=1}^n |A_{ij}|^2$$

This means we have

$$\text{trace} A^T A = \sum_{j=1}^n \|a_j\|^2 = \sum_{j=1}^n \sum_{i=1}^n |A_{ij}|^2 = \sum_{i=1}^n \sum_{j=1}^n |A_{ij}|^2$$

Hence, we have

$$\|A\|_F = \sqrt{\text{trace} A^T A} = \sqrt{\sum_{j=1}^n \sum_{i=1}^n |A_{ij}|^2} = \sqrt{\sum_{i=1}^n \sum_{j=1}^n |A_{ij}|^2}$$

b.

We want to show for $A, U, V \in \mathbb{R}^{n \times n}$ s.t. U, V are orthogonal that

$$\|UA\|_F = \|AV\|_F = \|A\|_F$$

First, we'll show $\|UA\|_F = \|A\|_F$. Recall that since U is orthogonal, we have $U^\top U = I$. Then we have

$$\|UA\|_F = \sqrt{\text{trace}(UA)^\top UA} = \sqrt{\text{trace}A^\top U^\top UA} = \sqrt{\text{trace}A^\top IA} = \sqrt{\text{trace}A^\top A} = \|A\|_F$$

Now, we'll show $\|AV\|_F = \|A\|_F$. Note that from part *a* we have for any matrix $B \in \mathbb{R}^{n \times n}$

$$\|B\|_F = \sqrt{\sum_{i=1}^n \sum_{j=1}^n |B_{ij}|^2} = \sqrt{\sum_{i=1}^n \sum_{j=1}^n |B_{ij}^\top|^2} = \|B^\top\|_F$$

Also recall that since V is orthogonal, we have $VV^\top = I$. Then we have

$$\begin{aligned} \|AV\|_F &= \|(AV)^\top\|_F = \|V^\top A^\top\|_F = \sqrt{\text{trace}(V^\top A^\top)^\top V^\top A^\top} = \sqrt{\text{trace}AVV^\top A^\top} = \sqrt{\text{trace}AIA^\top} = \sqrt{\text{trace}AA^\top} \\ &= \|A^\top\|_F = \|A\|_F \end{aligned}$$

Thus for $A, U, V \in \mathbb{R}^{n \times n}$ s.t. U, V are orthogonal, we have

$$\|UA\|_F = \|AV\|_F = \|A\|_F$$

c.

Let $A = U\Sigma V^\top$ be the full SVD of A . Then $U, \Sigma, V^\top \in \mathbb{R}^{n \times n}$ s.t. $U, V^\top$ are orthogonal. Recall from part *b* that the Frobenius norm is not changed by a pre- or post- orthogonal transformation. This means we have

$$\|A\|_F = \|U(\Sigma V^\top)\|_F = \|\Sigma V^\top\|_F = \|\Sigma\|_F$$

Now recall from part *a* that the Frobenius norm of a matrix is the square root of the sum of the squares of all the matrix entries. This means we have

$$\|A\|_F = \|\Sigma\|_F = \left\| \begin{bmatrix} \sigma_1 & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_r & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & 0 & \dots & 0 \end{bmatrix} \right\|_F = \sqrt{\sigma_1^2 + \sigma_2^2 + \dots + \sigma_r^2}$$

We've thus shown that $\|A\|_F = \sqrt{\sigma_1^2 + \sigma_2^2 + \dots + \sigma_r^2}$ where $\sigma_1, \sigma_2, \dots, \sigma_r$ are the singular values of A . By construction of the SVD, we have $\sigma_1 = \sigma_{\max}(A)$. Then we can rewrite $\|A\|_F$ as

$$\|A\|_F = \sqrt{\sigma_1^2 + \sigma_2^2 + \dots + \sigma_r^2} = \sqrt{\sigma_{\max}^2(A) + \sum_{i=2}^r \sigma_i^2}$$

Notice that for each σ_i , we have $0 \leq \sigma_i^2 \leq \sigma_{\max}^2(A)$. This means we have

$$\begin{aligned} \|A\|_F &= \sqrt{\sigma_{\max}^2(A) + \sum_{i=2}^r \sigma_i^2} \geq \sqrt{\sigma_{\max}^2(A) + \sum_{i=2}^r 0} = \sigma_{\max}(A) \\ \|A\|_F &= \sqrt{\sigma_{\max}^2(A) + \sum_{i=2}^r \sigma_i^2} \leq \sqrt{\sigma_{\max}^2(A) + \sum_{i=2}^r \sigma_{\max}^2(A)} = \sqrt{r} \sigma_{\max}(A) \end{aligned}$$

Therefore, we've shown

$$\sigma_{\max}(A) \leq \|A\|_F \leq \sqrt{r} \sigma_{\max}(A)$$

Finally, because the operator norm $\|A\| = \sigma_{\max}(A)$ and we proved $\sigma_{\max}(A) \leq \|A\|_F$, it follows that for every x ,

$$\|Ax\| \leq \|A\| \|x\| = \sigma_{\max}(A) \|x\| \leq \|A\|_F \|x\|$$

15.2350.)

a.

To do this problem, we'll first rewrite $\mathcal{E}_2$. Since A is invertible, we have

$$\begin{aligned} x = A^{-1}y &\Rightarrow \mathcal{E}_2 = \{y : \|A^{-1}y\| \leq 1\} \\ \|A^{-1}y\| \leq 1 &\Leftrightarrow \|A^{-1}y\|^2 \leq 1 \Leftrightarrow (A^{-1}y)^\top (A^{-1}y) \leq 1 \\ &\Leftrightarrow y^\top (AA^\top)^{-1}y \leq 1 \\ &\Rightarrow \mathcal{E}_2 = \{y : y^\top (AA^\top)^{-1}y \leq 1\} \end{aligned}$$

We're given S and want to choose A such that the following two sets are equal to each other

$$\mathcal{E}_1 = \{x : x^\top Sx \leq 1\} = \{y : y^\top (AA^\top)^{-1}y \leq 1\} = \mathcal{E}_2$$

Comparing the two, we see that we need

$$S^{-1} = AA^\top$$

Since $S^\top = S > 0$, we know that a unique positive definite square root $S^{1/2}$ exists and is invertible. We can use this to find a satisfactory choice of A

$$A = S^{-1/2} \Rightarrow AA^\top = S^{-1/2}(S^{-1/2})^\top = S^{-1/2}S^{-1/2} = S^{-1}$$

Thus, with $A = S^{-1/2}$, we have $\mathcal{E}_1 = \mathcal{E}_2$.

b.

We can reuse our work from part *a*. For $\mathcal{E}_1 = \mathcal{E}_2$ to hold, we need

$$S = (AA^\top)^{-1}$$

We're given A and want to choose S . Since A is invertible, we can select $S = (AA^\top)^{-1}$. Thus, with $S = (AA^\top)^{-1}$, we have $\mathcal{E}_1 = \mathcal{E}_2$.

c.

Now we want to expand on the uniqueness for our solutions for parts *a* and *b*. For part *b*, our solution is unique because the inverse of a matrix is unique, so $S = (AA^\top)^{-1}$ is the only choice. However, for part *a* our solution is not unique. Since $S = S^\top > 0$, we can rewrite S and S^{-1} in terms of their eigenvectors and eigenvalues,

$$\begin{aligned} S &= Q\Lambda Q^\top \\ \Rightarrow S^{-1} &= (Q\Lambda Q^\top)^{-1} = Q^{-\top}\Lambda^{-1}Q^{-1} = Q\Lambda^{-1}Q^\top \end{aligned}$$

We can also take the SVD of A ,

$$\begin{aligned} A &= U\Sigma V^\top \\ \Rightarrow A^\top &= (U\Sigma V^\top)^\top = V\Sigma^\top U^\top = V\Sigma U^\top \\ \Rightarrow AA^\top &= U\Sigma V^\top V\Sigma U^\top = U\Sigma^2 U^\top \end{aligned}$$

We can then rewrite our problem for part *a* to be given Q and Λ , choose $U, \Sigma, V^\top$ s.t.

$$Q\Lambda^{-1}Q^\top = U\Sigma^2 U^\top$$

This holds for when $U = Q$, $\Sigma = \Lambda^{-1/2}$, and $V^\top$ is orthogonal. Therefore, the solution for part *a* is not unique because V is free to be any orthogonal matrix. For instance, if $V^\top = Q^\top$, then

$$A = U\Sigma V^\top = Q\Lambda^{-1/2}Q^\top = S^{-1/2}$$

Is our choice of solution for part *a*. The produce of two orthogonal matrices is orthogonal, so an equivalent condition to $V^\top$ must be orthogonal is $V^\top = Q^\top X$ s.t. X is orthogonal. In other words, all solutions for A can be written as

$$A = U\Sigma V^\top = Q\Lambda^{-1/2}Q^\top X = S^{-1/2}X$$

This shows that the solution for part *a* is not unique: for any orthogonal matrix X , $A = S^{-1/2}X$ gives the same ellipsoid $\mathcal{E}_1 = \mathcal{E}_2$.

15.2470.)

a.

Given $y_1, \dots, y_N \in \mathbb{R}^n$, we want to find a hidden linear explanation for the data. We want to find m , A , b and $x_1, \dots, x_N \in \mathbb{R}^m$ and verify that

$$\begin{aligned} J &= \left(\frac{1}{N} \sum_{i=1}^N \|y_i - Ax_i - b\|^2 \right)^{1/2} \\ \frac{1}{N} \sum_{i=1}^N x_i &= 0 \\ \frac{1}{N} x_i x_i^\top &= I \end{aligned}$$

First, we'll find b . Since we want $y_i \approx Ax_i + b$, we have

$$\begin{aligned} y_1 + \dots + y_N &\approx Ax_1 + b + Ax_2 + b + \dots + Ax_N + b \\ \Rightarrow \frac{1}{N}(y_1 + \dots + y_N) &\approx \frac{1}{N}A(x_1 + \dots + x_N) + \frac{Nb}{N} \\ \Rightarrow \frac{1}{N}(y_1 + \dots + y_N) &\approx A\left(\frac{1}{N} \sum_{i=1}^N x_i\right) + b \\ \Rightarrow \frac{1}{N}(y_1 + \dots + y_N) &\approx b \end{aligned}$$

Therefore we choose $b = \frac{1}{N}(y_1 + \dots + y_N)$. For each y_i define $\tilde{y}_i = y_i - b$. Then we have

$$\tilde{y}_i = y_i - b \approx Ax_i$$

We can then group all of the x_i and $\tilde{y}_i$ together to make matrix multiplications. Define $X = [x_1 \ \dots \ x_N]$ and define $\tilde{Y} = [\tilde{y}_1 \ \dots \ \tilde{y}_N]$. Then we can rewrite the hidden linear explanation for the data

$$\tilde{Y} \approx AX$$

Let $\tilde{Y} = U\Sigma V^\top$ be the SVD of $\tilde{Y}$. Then we want to choose A and X such that

$$\begin{aligned} U\Sigma V^\top &\approx AX \\ \frac{1}{N} x_i x_i^\top &= I \Leftrightarrow \frac{1}{N} X X^\top = I \end{aligned}$$

Notice that if we set $X = V^\top$ then the second condition almost holds

$$\frac{1}{N} X X^\top = \frac{1}{N} (V^\top V) = \frac{1}{N} I$$

Therefore, suppose

$$X = \sqrt{N} V^\top$$

so that

$$\frac{1}{N} X X^\top = \frac{1}{N} (\sqrt{N} V^\top) (\sqrt{N} V^\top)^\top = V^\top V = I$$

which satisfies the unit covariance condition. Then, to satisfy $\tilde{Y} \approx AX$, we would have

$$A = \frac{1}{\sqrt{N}} U \Sigma$$

With these choices, we have

$$\tilde{Y} \approx AX = \frac{1}{\sqrt{N}} U \Sigma \cdot \sqrt{N} V^\top = U \Sigma V^\top$$

so the approximation holds exactly if we use all singular values. To choose a smaller number of factors $m < n$, we examine the singular values in Σ . Large singular values have high impact on the output, while very small singular values contribute little. A reasonable choice of m is to keep the top m singular values that provide impact above a certain threshold, say 95%. For example, if we have

$$\sigma_1 = 10, \quad \sigma_2 = 4, \quad \sigma_3 = 0.1, \quad \sigma_4 = 0.05,$$

then we might choose $m = 2$. We can formalize this by finding the minimum m such that

$$\frac{\sum_{i=1}^m \sigma_i}{\sum_{j=1}^n \sigma_j} \geq \text{threshold} = 0.95$$

Once m is chosen, we modify $U, \Sigma, V^\top$ to be $\hat{U}, \hat{\Sigma}, \hat{V}^\top$ s.t. the modified SVD matrices keep only the top m singular values and corresponding singular vectors:

$$\begin{aligned} U \Sigma V^\top &= \begin{bmatrix} u_1 & u_2 & \cdots & u_n \end{bmatrix} \begin{bmatrix} \sigma_1 & 0 & \cdots & 0 & 0 & \cdots & 0 & 0 \\ 0 & \sigma_2 & \cdots & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & \sigma_n & 0 & \cdots & 0 & 0 \end{bmatrix} \begin{bmatrix} v_{11}^\top & v_{12}^\top & \cdots & v_{1N}^\top \\ v_{21}^\top & v_{22}^\top & \cdots & v_{2N}^\top \\ \vdots & \vdots & \ddots & \vdots \\ v_{N1}^\top & v_{N2}^\top & \cdots & v_{NN}^\top \end{bmatrix} = \sum_{i=1}^n \sigma_i u_i v_i^\top \\ \tilde{Y} &\approx \sum_{i=1}^m \sigma_i u_i v_i^\top = \begin{bmatrix} u_1 & u_2 & \cdots & u_m & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} \sigma_1 & 0 & \cdots & 0 & 0 & \cdots & 0 & 0 \\ 0 & \sigma_2 & \cdots & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & \sigma_m & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & 0 & \cdots & 0 & 0 \end{bmatrix} \begin{bmatrix} v_{11}^\top & v_{12}^\top & \cdots & v_{1N}^\top \\ v_{21}^\top & v_{22}^\top & \cdots & v_{2N}^\top \\ \vdots & \vdots & \ddots & \vdots \\ v_{m1}^\top & v_{m2}^\top & \cdots & v_{mN}^\top \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix} \\ \hat{U} &= \begin{bmatrix} u_1 & u_2 & \cdots & u_m \end{bmatrix} \in \mathbb{R}^{n \times m} \\ \hat{\Sigma} &= \begin{bmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_m \end{bmatrix} \in \mathbb{R}^{m \times m} \\ \hat{V}^\top &= \begin{bmatrix} v_{11}^\top & v_{12}^\top & \cdots & v_{1N}^\top \\ v_{21}^\top & v_{22}^\top & \cdots & v_{2N}^\top \\ \vdots & \vdots & \ddots & \vdots \\ v_{m1}^\top & v_{m2}^\top & \cdots & v_{mN}^\top \end{bmatrix} \in \mathbb{R}^{m \times N} \end{aligned}$$

Finally, we choose

$$X = \sqrt{N} \hat{V}^\top, \quad A = \frac{1}{\sqrt{N}} \hat{U} \hat{\Sigma}$$

We've thus found a hidden linear explanation, $m, A, b, x_1, \dots, x_N$ of the data $y_1, \dots, y_N$. Now we just need to verify

$$\frac{1}{N} \sum_{i=1}^N x_i = 0, \quad \frac{1}{N} \sum_{i=1}^N x_i x_i^\top = I.$$

Since we defined $\tilde{Y} = Y - b$, the data has been centered, i.e.,

$$\sum_{i=1}^N \tilde{y}_i = \sum_{i=1}^N y_i - b = \left(\sum_{i=1}^N y_i \right) - N \left(\frac{1}{N} \sum_{i=1}^N y_i \right) = 0$$

We have $X = \sqrt{N} \hat{V}^\top$, where $\hat{V}$ contains the top m right singular vectors of $\tilde{Y}$. By properties of the SVD, these vectors lie in the space orthogonal to the mean of the columns of $\tilde{Y}$. Therefore,

$$\sum_{i=1}^N x_i = \sqrt{N} \sum_{i=1}^N \hat{v}_i^\top = 0 \Rightarrow \frac{1}{N} \sum_{i=1}^N x_i = 0.$$

By construction, the columns of $\hat{V}$ are orthonormal, so

$$\frac{1}{N} \sum_{i=1}^N x_i x_i^\top = \frac{1}{N} X X^\top = \frac{1}{N} (\sqrt{N} \hat{V}^\top) (\sqrt{N} \hat{V}^\top)^\top = \hat{V}^\top \hat{V} = I_m.$$

Hence, the factors $x_1, \dots, x_N$ satisfy the required conditions.

b.

This question was done in Julia. The code and plot can be seen below. The verification for the required conditions for the x vectors can be seen in the output in the terminal below the code.

```

1  using LinearAlgebra, Plots, Statistics
2
3  #Question 4.)
4
5  include("readclassjson.jl")
6  data = readclassjson("linexp_data.json")
7
8  Y = data["Y"]
9  n, N = size(Y)
10
11 #Choose b
12 b = sum(Y, dims=2) / N
13 tildeY = Y .- b
14
15 #Choose m
16 U, Sigma, V = svd(tildeY; full=false)
17 threshold = 0.95
18 cumsum_sv = cumsum(Sigma) / sum(Sigma)
19 m = findfirst(x -> x >= threshold, cumsum_sv)
20
21 # Truncate SVD
22 Uhat = U[:, 1:m]
23 Sigmahat = Diagonal(Sigma[1:m])
24 Vhat = V[:, 1:m]
25
26 #Choose X and A
27 X = sqrt(N) * Vhat'
28 A = (1/sqrt(N)) * Uhat * Sigmahat
29
30 #Compute + Sort + Plot Errors
31 errors = [norm(Y[:,i] - A*X[:,i] - b) for i in 1:N]
32 sorted_errors = sort(errors, rev=true)
33 plot(sorted_errors, xlabel="Sorted Index", ylabel="Norm of Error Vector",
34      title="Distribution of Explanation Errors", legend=false)
35 savefig("hw7_q4b.png")
36
37 #Verify X vector properties
38 mean_X = sum(X, dims=2) / N
39 cov_X = (X * X') / N
40
41 println("Mean of x vectors (should be 0):")
42 println(mean_X)
43 println("m = ", m)
44 println("\nCovariance of x vectors (should be I_m):")
45 println(cov_X)

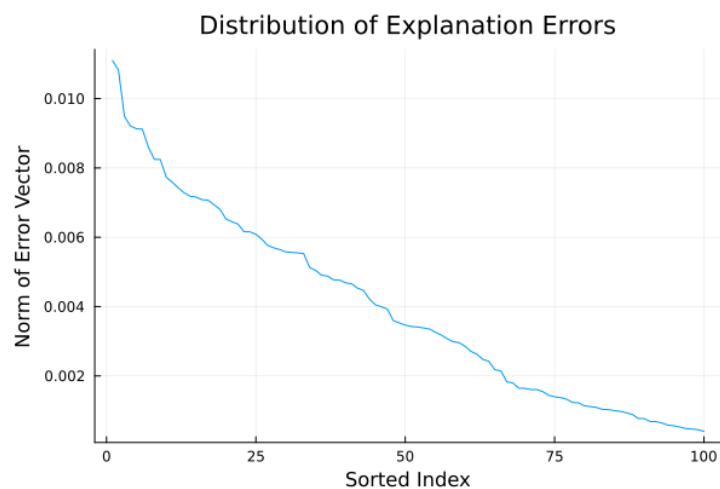
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```

Mean of x vectors (should be 0):
[1.038958238045214e-16; -1.450228821916106e-17; 4.285408875013384e-16;]
m = 3

Covariance of x vectors (should be I_m):
[1.0000000000000002; -8.46072615417012e-17; 9.9400713130277e-20; -8.46072615417012e-17; 1.0000000000000001; 1.662121654895695e-16; 9.9400713130277e-20; 1.662121654895695e-16; 1.0000000000000001]

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16.2570.)

a.

We want to explain how to project to the closest matrix satisfying the rank constraint and project to the closest matrix with the known entries. For the first projection, we want $\hat{A}^{(k)}$ to be the matrix closest to $\tilde{A}^{(k)}$ in Frobenius norm s.t. $\text{rank}(\hat{A}^{(k)}) \leq r$. To find the projection, first find the SVD

$$\tilde{A}^{(k)} = U \Sigma V^\top = \sum_{i=1}^q \sigma_i u_i v_i^\top$$

Where $q = \text{rank}(\tilde{A}^{(k)})$. Then the optimal rank r approximator is

$$\hat{A}^{(k)} = \sum_{i=1}^r \sigma_i u_i v_i^\top$$

This is because σ_i are ordered in terms of importance of impact to matrix A , $\sigma_1 \geq \sigma_2 \geq \dots$, and this rank r approximator minimizes the operator norm of the difference matrix of $\hat{A}^{(k)}$ with $\tilde{A}^{(k)}$,

$$\|\hat{A}^{(k)} - \tilde{A}^{(k)}\| = \sigma_{p+1}$$

Any other rank r matrix would have a difference matrix operator norm $\geq \sigma_{p+1}$, and this approximator also minimizes the Frobenius norm of the difference matrix. For the second projection, we want $\hat{A}^{(k+1)}$ to be the matrix closest to $\tilde{A}^{(k)}$ in Frobenius norm s.t. $\hat{A}_{ij}^{(k+1)} = A_{ij}^{\text{known}}$ for $(i, j) \in \mathcal{K}$. This projection is simple. We can define

$$\hat{A}_{ij}^{(k+1)} = \begin{cases} A_{ij}^{\text{known}} & \text{if } (i, j) \in \mathcal{K} \\ \tilde{A}_{ij}^{(k)} & \text{if } (i, j) \notin \mathcal{K} \end{cases}$$

Then we have

$$\begin{aligned} \|\hat{A}^{(k+1)} - \tilde{A}^{(k)}\| &= \left(\sum_{(i,j) \in \mathcal{K}} (\hat{A}_{ij}^{(k+1)} - \tilde{A}_{ij}^{(k)})^2 + \sum_{(i,j) \notin \mathcal{K}} (\hat{A}_{ij}^{(k+1)} - \tilde{A}_{ij}^{(k)})^2 \right)^{\frac{1}{2}} \\ &= \left(\sum_{(i,j) \in \mathcal{K}} (A_{ij}^{\text{known}} - \tilde{A}_{ij}^{(k)})^2 + \sum_{(i,j) \notin \mathcal{K}} (\tilde{A}_{ij}^{(k)} - \tilde{A}_{ij}^{(k)})^2 \right)^{\frac{1}{2}} = \left(\sum_{(i,j) \in \mathcal{K}} (A_{ij}^{\text{known}} - \tilde{A}_{ij}^{(k)})^2 \right)^{\frac{1}{2}} \end{aligned}$$

As $\hat{A}^{(k+1)}$ is subject to $\hat{A}_{ij}^{(k+1)} = A_{ij}^{\text{known}}$ for $(i, j) \in \mathcal{K}$, the Frobenius cost between $\hat{A}^{(k+1)}$ and $\tilde{A}^{(k)}$ must be at least $\left(\sum_{(i,j) \in \mathcal{K}} (A_{ij}^{\text{known}} - \tilde{A}_{ij}^{(k)})^2 \right)^{\frac{1}{2}}$. Therefore, our definition of $\hat{A}^{(k+1)}$ is the optimal projection.

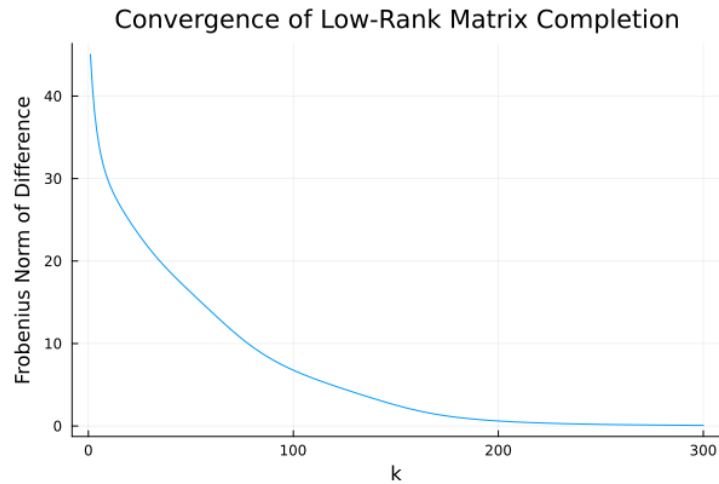
b.

This question was done in Julia. The code, plot, and comment can be seen below.

```

1 using LinearAlgebra, Plots, Statistics
2
3 #Question 5.)
4
5 include("readclassjson.jl")
6 data = readclassjson("low_rank_matrix_completion.json")
7
8 m = data["m"]
9 n = data["n"]
10 r = data["r"]
11 Atrue = data["Atrue"]
12 Aknown = data["Aknown"]
13 K = data["K"]
14 p = data["p"]
15
16 Ahat = fill(mean(Aknown), m, n)
17 for i in 1:p
18     Ahat[K[i, 1], K[i, 2]] = Aknown[i]
19 end
20 error = []
21 for i in 1:300
22     F = svd(Ahat)
23     Atilde = F.U[:, 1:r] * Diagonal(F.S[1:r]) * F.Vt[1:r, :]
24     global Ahat = Atilde
25     for j in 1:p
26         global Ahat[K[j, 1], K[j, 2]] = Aknown[j]
27     end
28     push!(error, norm(Ahat - Atrue))
29 end
30
31 p_tradeoff = plot(1:300, error, xlabel="k",
32                  ylabel="Frobenius Norm of Difference",
33                  title="Convergence of Low-Rank Matrix Completion",
34                  label="")
35 savefig("hw7_q5b.png")

```



As can be seen in the plot above, the algorithm performed very well for the given data set. The error decreases rapidly at the start and then plateaus around the 200th iteration of the algorithm. Since we were only given roughly one-sixth of the known entries, and the final error was

$$\|\hat{A}^{(300)} - A\|_F = 0.077$$

This indicates that the algorithm approximated the underlying low-rank structure with high accuracy despite the limited data.